

It is known that the fundamental group of the boundary  $\partial V$  of the solid torus  $V$  is the free abelian group on two generators  $\mathbb{Z} \times \mathbb{Z}$ . Therefore one can define a longitude for  $V$  as a simple, closed curve on  $\partial V$  that is of the form  $(k, 1)$  in the fundamental group of the boundary  $\partial V$ ,  $k \in \mathbb{Z}$ . A meridian is a simple, closed curve of the form  $(1, 0)$ .

**Definition 3.1.** Let  $D$  be a diagram of an oriented link  $L$  and let  $c_1, c_2$  be two components of  $D$ . The *linking number* between  $c_1$  and  $c_2$  is the total number of positive crossings minus the total number of negative crossings, divided by 2, over all crossings between  $c_1$  and  $c_2$ . A *positive* crossing is when one has to rotate the over arc counterclockwise, in order to align the two arcs involved, so that they both point to the same direction. A *negative* crossing is when the upper strand has to rotate clockwise. The linking number counts the algebraic number of times one component winds around the other one. Further, it is an ambient isotopy invariant as it respects Reidemeister II and III moves and it is not affected by the Reidemeister I move. Thus we talk about the *linking number of two link components* as the linking number between them in any diagram representative of the link.

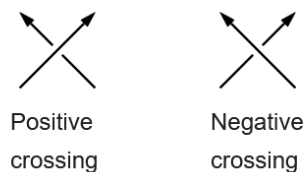


Figure 5: The two types of crossing in a link diagram.

**Definition 3.2** (Framing). Let  $K$  be an oriented knot in  $S^3$  (or  $\mathbb{R}^3$ ). Then  $K$  has a regular neighbourhood in  $S^3$  that is a solid torus  $S^1 \times D^2 = V$ , where  $K$  is the core of  $V$ . Let  $K_0$  be a longitude in  $V$  with the same orientation as  $K$ . The linking number  $lk(K, K_0)$  between  $K$  and  $K_0$ , is called a *framing* of  $K$ . For an example view Fig. 6a. A *framed link* of  $c$ -components is a link of whom each one of the  $c$  components is a framed knot. To a framed link of  $c$  components one can assign a framing vector  $\vec{\lambda} = (\lambda_1, \dots, \lambda_c)$ , where  $\lambda_i \in \mathbb{Z}$  is the framing of the  $i$ -th component.

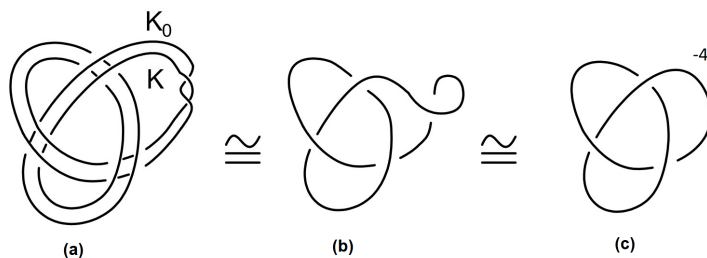


Figure 6: Different framing representations of a trefoil with framing -4.

We note that a framing for a knot can be any integer, as it also depends on how many times the longitude follows a meridian of  $V$ , which is the first coordinate in the fundamental group  $\mathbb{Z} \times \mathbb{Z}$  of the torus  $\partial V$ .